

THE AI TOOLBOX · CLASS 01

What Is AI? What Can AI Actually Do?

From science fiction to reality — understanding AI, machine learning & deep learning.

Merit Point Academy · 60 min · press **S** notes · → next ·  to comment

YOUR INSTRUCTOR

Bo Shang (商博)

Postdoctoral Scholar, AI & Mobility Research Lab, CUNY City College of New York.

- **PhD**, Pattern Recognition & Intelligent Systems — Northeastern University (China), with an exchange PhD stint at **UC Merced**.
- Research: **robotics & autonomous systems**, LiDAR–camera sensor fusion, computer vision, bridge-inspection automation, traffic monitoring, unmanned aerial systems.
- Previously taught/researched at UC Merced, Missouri S&T, Vaughn College, and CCNY before joining Merit Point Academy (恒点学术).



Why this matters to you:

the autonomous-vehicle perception example running through this course is my own research world — not a textbook toy.

My teaching approach: every concept lands on something you build and run — I'd rather you leave with one working result than ten memorized definitions.

15 classes, one research pipeline

Every class delivers **one tangible result + one research skill + one AI-tool workout**, building toward one arc: **idea** → **data** → **model** → **experiment** → **presentation**.

Stage 1–2 · Classes 1–6

AI literacy, finding a research question, and the AI toolbox (ChatGPT, Elicit, Semantic Scholar, Kaggle).

Stage 3 · Classes 7–9

Data science: finding, cleaning, and visualizing real datasets in Python/Colab.

Stage 4–5 · Classes 10–15

ML models (regression → trees → neural nets), a fair experiment, and a final mini-study — paper + slides.

Anchor case: every data/model class also lands on one running example — teaching a car to see (autonomous-vehicle perception) — so ideas are never just abstract. Prefer a different domain? Swap it in — the case study is the example, not the requirement.

HOW YOU'LL BE ASSESSED

Assignments & grading — project-based, no exams

35% — Weekly tangible results (your growing research portfolio)

20% — Data work: selection, cleaning & visualization

25% — Modeling & experiment: fair comparison, honest metrics

20% — Final mini-study: paper + presentation

Every class ends with homework before the next session — usually finishing that day's tangible result, plus an optional "stretch" task. Bring questions to class or post them with .

Using AI honestly: brainstorm, search, and debug with AI freely — but verify every claim and citation, never paste private data into AI tools, and disclose how you used AI in your work.

JOIN OUR GOOGLE CLASSROOM

Where slides, homework & grades live



Class code

kday73zp

classroom.google.com → tap + → **Join class** → enter the code above.



What's posted there

- Every class's slide deck, under **Classwork** → **Class Materials**
- Every homework assignment, under **Classwork** → **Assignments** — submit your work (Colab link, doc, or file) there
- Grades & feedback, under **Grades** — this is also where you'll see graded homework

Join before next class so you can submit today's homework there. Questions about Classroom itself → ask your instructor, not  on a slide.

AGENDA · HOW THIS HOUR RUNS

Where we're going

- **0–10'** Warm-up + a 70-year history of AI
- **10–22'** AI on screen: traditional → current → future
- **22–34'** AI $\supset$ ML $\supset$ DL, and where AI hides
- **34–42'** AI's application map + AI as a research tool
- **42–50'** Live demos: ChatGPT · Perplexity · ScholarMind
- **50–58'** Two hands-on Python labs (with an AI helper)
- **58–60'** Project, homework & wrap-up
- Quizzes &  comments throughout

By the end: tell AI/ML/DL apart, name AI in a field you love, and explain why AI is an instrument for research — not just a product.

WARM-UP · ACTIVATE

Quick check · tap an answer

You woke up, unlocked your phone with your face, and your music app suggested a song. How many AI systems was that?

Zero — those are just normal phone features

One — only the face unlock

At least two — face unlock and the recommendation

None of these count unless a robot is involved

A summer that became 70 years

In 1956, ten scientists gathered at Dartmouth for what they expected to be a single summer's work: build a machine that could **"use language, form abstractions, and solve problems reserved for humans."** They coined the name **"artificial intelligence"** — and wildly underestimated the problem. Three years later, Arthur Samuel wrote a checkers program that got better the more it played against itself. He called what it did **"machine learning."**

[Speak this quote](#)

Neural voice · click to play / pause



The founders thought intelligence was a summer project. Samuel's self-improving checkers player was the first hint: don't *program* intelligence — let a machine *learn* from examples.

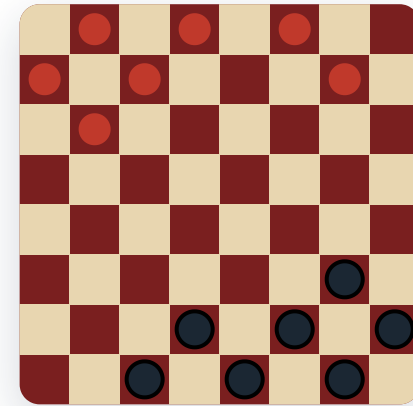
The program that taught itself checkers

Arthur Samuel, an engineer at **IBM**, built a checkers (draughts) program on the room-sized **IBM 701** — and did something radical: he let it **improve on its own**.

- It played **thousands of games against itself**, remembering which positions led to wins.
- It tuned its own scoring of board positions — an early ancestor of today's **reinforcement learning**.
- Samuel coined the term "**machine learning**" to describe it (1959).
- By **1962** it beat a strong human amateur — an early, very public win for AI.

► Watch: Samuel's checkers on the IBM 701

Why it mattered: the first convincing proof a machine could get better from **experience**, not just follow fixed rules. That single idea — learn from data — is the seed of every system in this course.



 learns by self-play



scan / tap to open

70 years of winters and springs

AI didn't arrive overnight. It boomed, froze, and boomed again — each cycle banking hard-won lessons.

1956

Dartmouth meeting

The term "artificial intelligence" is coined — the field is born.

1970s

First AI winter

Too little compute and data; early systems stall, funding dries up.

1980s

Expert systems

Rule-based systems find real business use — a short, bright spring.

1990s

Second AI winter

Expert systems are costly and brittle; enthusiasm cools again.

2010s+

Big data + deep learning

Compute + huge datasets ignite neural nets — today's golden era.

Knowing this history keeps you **level-headed** about hype: every boom looked unstoppable, and every winter looked permanent.
Today is a real boom — but ask what's hype.

HOOK · THINK-PAIR-SHARE

Which of these is AI?

Face unlock · autocomplete · a thermostat · a chess app · a calculator —
which ones count as "AI", and **how can you tell?**

Face unlock

Autocomplete

Thermostat

Chess app

Calculator

Hold your answer — the next three clips give you a rule for deciding.

AI ON SCREEN · TRADITIONAL AI · ~1950-2000

Deep Blue beats Kasparov (1997)

FEATURE: SEARCH + HAND-WRITTEN RULES

A chess machine that **never learned** — it searched ~200 million positions per second using rules experts wrote by hand. This is "**good old-fashioned AI**": smart because humans coded the cleverness in.

▶ Watch the clip

DEEP
BLUE



scan / tap to open

AI ON SCREEN · CURRENT AI · 2012–NOW

ChatGPT writes, explains, and codes

FEATURE: LEARNING FROM DATA (MACHINE LEARNING)

Today's AI **learns patterns from oceans of examples** instead of following hand-written rules. One model can chat, summarize, translate, and code — because it learned language, not chess. **This is machine learning at scale.**

▶ Watch the clip

Chat
GPT



scan / tap to open

“Her” — a glimpse of AGI

FEATURE: GENERAL, AUTONOMOUS, HUMAN-LEVEL (AGI)

Sci-fi like *Her* imagines **Artificial General Intelligence**: one system that learns **any** task, sets its own goals, and reasons across everything. **It doesn't exist yet** — separating real from movie is itself a research skill.

Film	Year	Rating	AGI angle
<i>Her</i>	2013	R	Emotionally intelligent AGI that learns, grows, then surpasses human cognition
2001: A Space Odyssey	1968	G	HAL 9000 — an AI that reasons, then turns against its crew
I, Robot	2004	PG-13	Robots developing free will and emergent consciousness
The Imitation Game	2014	PG-13	Turing & the foundational question: can machines think?
WALL·E	2008	G	Gentle machine consciousness and emotion
Ex Machina	2014	R	The Turing test dramatized; AI deception and self-preservation
Blade Runner 2049	2017	R	Synthetic beings, identity, and what counts as "alive"

G **PG-13** = screen clips in class · **R** = mature themes, optional further watching

CHECK · AFTER THE CLIPS

Quick check · tap an answer

Deep Blue searched millions of chess moves with rules experts wrote by hand. Which kind of AI is that?

Modern machine learning — it learned from data

Traditional AI — search and hand-written rules

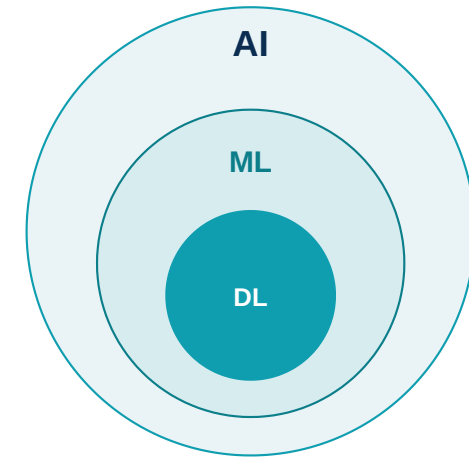
AGI — general human-level intelligence

It isn't AI at all

CONCEPT 1

AI $\supset$ Machine Learning $\supset$ Deep Learning

- **AI** — the whole field of machine intelligence
- **ML** — systems that improve from *data*, not hand-written rules
- **DL** — machine learning using many-layered neural networks



KEY IDEA — When people say "AI" today, they usually mean **machine learning**. Deep Blue was AI but not ML; ChatGPT is all three.

CONCEPT 2 · WHAT POWERS EACH LAYER

Different tools at each ring

AI

Broadest vision. Less data-driven. Key tech: **expert systems, search algorithms.** (Deep Blue lives here.)

ML

Learns from data. Tunes a model's parameters. Key tech: **decision trees, SVMs.** Used for prediction & classification.

DL

Neural networks. Needs big data + compute. Key tech: **CNNs, RNNs, Transformers.** Powers vision & language.

AI is the grand vision · ML is the path that got us there · DL is ML's strongest tool for hard, messy data.

CONCEPT 3

AI is already everywhere

You use a dozen AI systems before lunch. Naming them is the first research skill — seeing the AI hiding in plain sight.



Recommendation feeds



Spam filters



Navigation & maps



Face unlock



Voice assistants



Photo search & tagging

Each one is a **model trained on data** — not a person, not magic, not hand-coded rules.

Quick check · tap an answer

Which statement is TRUE about AI, ML, and DL?

They all mean the same thing

Every machine-learning system is AI, but not every AI is machine learning

Deep learning is bigger than machine learning

AI is a kind of deep learning

Where AI already works



Medicine

Reads CT/MRI scans for tumors; **Google Health** matched experts at breast-cancer screening.



Biology

AlphaFold predicted 98.5% of human protein structures — a 50-year problem.



Climate

Forecasts extreme weather, models warming scenarios, optimizes renewable grids.



Finance

High-frequency trading, credit scoring, and real-time fraud detection.



Art

Midjourney & DALL·E turn text into images; AI composes music and drafts scripts.



Mobility

Self-driving perception — our running example for the whole course.

Same core idea — **learn patterns from data** — reshaped a dozen industries. Which field is yours?

AlphaFold: a 50-year problem, solved

Proteins fold into 3-D shapes that decide what they do. Predicting that shape stumped biology for **50 years**.

- DeepMind's **AlphaFold** predicted structures at near-atomic accuracy
- Covered **98.5%** of all human proteins
- Freely released — now used by labs worldwide for disease & drug research



Sequence → 3-D structure

What took months in a lab now takes seconds.

This is AI as an **instrument of discovery** — not a chatbot, but a microscope for life itself.

CONCEPT 4

AI as your research super-assistant

Beyond products, AI is an **instrument**. It speeds up every stage of research.



Literature review

Scan and summarize hundreds of papers; map a field fast.



Hypothesis ideas

Surface hidden links across data to suggest what to test.



Data analysis

Crunch huge, high-dimensional data; spot patterns & outliers.



Code help

Write, explain, and debug research code — even for non-programmers.

Next: a concrete workflow that turns a vague topic into a finished, reviewed PDF.

From a topic to a reviewed PDF



Steps **6** → **7** → **8** form a **review loop** — repeat *review* → *revise* → *update* until it's good. The human stays in charge; AI does the heavy lifting.

WHAT YOU ACTUALLY LEARN

Two skills this workflow trains

The Skill of Asking

Choosing the right question

Picking and sharpening the **right question** (steps 1–2). A good question is specific, measurable, and answerable — it decides everything downstream.

The Skill of Judging

Telling good work from weak

Reading a draft (steps 6–8) and saying precisely what's strong, what's wrong, and what to fix — **taste you can defend**.

AI can draft for you. It **cannot** ask your question or judge your taste — those stay human, and they're what we grade.

Quick check · tap an answer

In the research workflow, what do steps 6 → 7 → 8 (review → revise → update) form?

A one-time final step

A review loop you repeat until the work is good

The very first thing you do

A part only the AI controls

LIVE DEMO 1

ChatGPT — your thinking partner

A large language model that drafts, explains, and brainstorms. Watch it help — then watch us check it.

You ▶ Explain AI vs. machine learning vs. deep learning to a 10-year-old, with one example each.

You ▶ I'm interested in self-driving cars. Give me 3 research questions a high-schooler could test.

You ▶ List 3 specific ways AI is used in medicine — name the system, its input, and its output.

You ▶ What's the capital of Australia — and how confident are you?



chatgpt.com

LIVE DEMO 2

Perplexity — answers with sources

Like ChatGPT, but it searches the web and shows where each claim came from. Use it to fact-check.

You ▶ What's the capital of Australia – and cite a source.

You ▶ How is AI used in climate science? Give 3 examples with links.

You ▶ Did a New York lawyer really get sanctioned for citing fake AI cases? Source it.

You ▶ Latest accuracy of AI at detecting pedestrians at night – with citations.



perplexity.ai

LIVE DEMO 3

ScholarMind AI — topic → research brief

Our research platform. It turns a rough interest into a sharp question and a downloadable brief (steps 1–3 of the workflow).

You ▶ Topic: 'I like animals and cameras.' → propose 3 testable research questions.

You ▶ Analyze: 'Can a model detect birds from photos?' – what data, model, and metric?

You ▶ Generate a research brief I can hand to a coding agent.

You ▶ Download brief.md and open it in Claude / Codex.



academic.hopeembark.org

Find a pattern in data — like a data scientist

On the Titanic, who was more likely to survive? Let's let the **data** answer. Edit the code, hit Run, and ask the assistant anything.

```
1 # Titanic survivors – a tiny real dataset
2 # Each person: (sex, passenger_class, survived?)
3 people = [
4     ("female", 1, 1), ("female", 1, 1), ("female", 3, 0),
5     ("male", 1, 0), ("male", 3, 0), ("female", 2, 1),
6     ("male", 2, 0), ("female", 3, 1), ("male", 1, 1),
7     ("female", 2, 1), ("male", 3, 0), ("female", 1, 1),
8 ]
9
10 def survival_rate(group):
11     members = [p for p in people if group(p)]
12     if not members:
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.

Watch a model learn from data (numpy)

This tiny model **learns** a rule from examples — no hand-written if-statements. It improves each step, exactly like real machine learning. Run it, then ask the assistant to explain how it learns.

```
1 import numpy as np
2
3 # Data: hours studied -> passed the test? (0/1)
4 hours = np.array([1, 2, 3, 4, 5, 6, 7, 8], dtype=float)
5 passed = np.array([0, 0, 0, 1, 1, 1, 1, 1], dtype=float)
6
7 # A simple model: probability = sigmoid(w*hours + b)
8 def sigmoid(z): return 1 / (1 + np.exp(-z))
9
10 w, b = 0.0, 0.0
11 lr = 0.1
12 for step in range(2000):
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.

Teaching a car to see

Autonomous driving threads the whole course. The car's AI splits into three jobs — and this course lives in the first one.

1 · Perception

See the world — detect pedestrians, cars, signs from camera images. (*CNNs*)

2 · Prediction

Forecast where each object moves next. (*RNNs / Transformers*)

3 · Planning

Decide: brake, steer, or keep going. (*reinforcement learning*)

Perception is the foundation: if the car sees wrong, perfect prediction and planning can't keep it safe.

DISCUSS · 3 MINUTES · PICK A SIDE

AI: more opportunity, or more risk?



The upside

Frees us from repetitive work, supercharges science, and personalizes learning, health, and services.



The risks

Job disruption, privacy loss, algorithmic bias, and misuse of powerful tools.

Turn to a neighbor: where does the balance tip for **you** — and why? Drop your take in .

YOUR PROJECT

Build an AI application map

Pick one domain you care about and map where AI already works in it — the first move of any researcher entering a field.

- Choose a domain (medicine, sport, music, climate...)
- Use ChatGPT + Perplexity to find 3 real systems
- Sketch a mind map linking them

Deliverable

A one-page mind map + 3 captioned cases.

Success looks like

3 specific examples (not "AI helps doctors"), each labeled with its **input** and **output**.

BUILD IT NOW · HANDS-ON

Map it with ChatGPT + Perplexity

1. Open ChatGPT and Perplexity
2. Ask: "How is AI used in <your domain>? Give 3 specific systems with sources."
3. Verify each one is real — open a source
4. Draw the mind map (paper, slides, or a whiteboard tool)

ChatGPT

Perplexity AI

ScholarMind

Tip: Perplexity shows its sources — use it to fact-check ChatGPT. That's the Skill of Judging in action.

Today's vocabulary

AI — machines doing tasks that normally need human intelligence.

Deep Learning — ML with many-layered neural networks for complex data.

Prediction — using a trained model to estimate future or unknown values.

Machine Learning — systems that learn patterns from data instead of explicit rules.

Perception — turning sensor input (images, sound) into understanding.

Planning — choosing a sequence of actions to reach a goal.

RECOMMENDED HOMEWORK

Before next class

1. **Finish your AI application map** — 3 real cases, each with input → output. Be ready to share.
2. **Explore one field** you love: find a recent AI use and note what problem it solves.
3. **Re-run a lab** from today and change one thing — bring a question for the AI assistant.

★ Stretch (optional)

Open Kaggle's classic "**Titanic — Machine Learning from Disaster**".
Skim how others predict who survived: data → features → model → score.

[kaggle.com · Titanic](https://www.kaggle.com/titanic)

Bring curiosity, not perfection. Post questions on any slide with 💬 — your instructor will see them.

TODAY'S LINE TO REMEMBER

"The best way to predict the future is to invent it."

— Alan Kay

Use AI to **create** the future, not just to predict it.



Speak this

WRAP-UP

Three things to keep

- AI $\supset$ ML $\supset$ DL — nested, not equal
- AI is already all around you — and across every science
- AI is an **instrument** for research — you own the question & the judgement

Exit ticket: name one AI system you used today, and guess what data trained it. Post it in .

Next · Class 2 — turn a vague interest into a real research question.

BACKUP MATERIAL

Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

AV perception, in depth

the tech behind seeing

AI in art

creativity & controversy

Bonus quiz

one more check

Inside the car's three jobs

Perception

Computer vision with CNNs extracts features from camera frames to detect pedestrians, lanes, and signs.

Prediction

RNNs / Transformers model time: where will that cyclist be in 2 seconds?

Planning

Reinforcement learning searches for a safe, comfortable path — when to brake, steer, go.

Perceive → Predict → Plan is a closed loop running many times per second.

AI in art — tool or artist?



Images

Midjourney & DALL·E turn a sentence into a picture — no drawing skill needed.



Music

AI composes melodies, arranges parts, even mimics a singer's voice.



Writing

Helps brainstorm plots, characters, and copy — a partner for writers.

Open question: when an algorithm creates, who is the **artist** — and what is "creativity"?

BACKUP · BONUS CHECK

Quick check · tap an answer

AlphaFold predicting protein structures is an example of AI as...

A chatbot for fun

An instrument that accelerates scientific discovery

Traditional rule-based AI

Artificial general intelligence